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Multimodal Transportation Route Optimization of Cold Chain Container in Time-Varying Network Considering Carbon Emissions

Shouchen Liu [1,2]

[1] School of Business Administration, Fujian Business University, Fuzhou 350012, China; 2019019082300004@ecjtu.edu.cn
[2] School of Transportation Engineering, East China Jiao Tong University, Nanchang 330013, China

Abstract: To solve the multimodal transport route optimization problem considering carbon emission, the vehicle speed has time-varying characteristics, and the customer has a time window limit. The carbon emission of multimodal transport system is affected by the energy consumption of transport vehicles in the time-varying network. The time-varying network is uncertain, and carbon emissions may continue to rise after a gradual decline. Based on this, this study established the sum of the carbon emission cost, transportation cost, penalty cost for exceeding the time window, and the damage cost of the cold chain cargo as the objective function. A route optimization model of cold chain container multimodal transportation was established. Static and dynamic optimization scenarios were designed and a hummingbird evolutionary genetic algorithm was used to solve the model. The effectiveness of the model and the practical value of the study are verified by the empirical analysis of the multimodal transport network of the Yangtze River Delta economic group. Results show that the dynamic model of the time-varying network can more truly reflect the transportation of the multimodal transport network and meet the efficiency requirements for the cold chain container multimodal transport. This study aims to solve the time-varying network under cold chain route optimization of container intermodal transportation, provides new insights for related businesses and a theoretical basis for reasonable multimodal transport route decisions.

Keywords: transportation; time-varying network; cold chain; improved queen bee genetic algorithm; carbon emissions

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1. Introduction

In 2021, the total demand of China's agricultural chain logistics was 295.6 million tons, 36.9 million tons more than the volume in 2020, and representing a year-on-year growth of 19.55%. Nearly 90% of the freight volume in cold chain transportation is completed by road transportation. Road cold chain transportation is undoubtedly the core and dominant mode in the cold chain transportation market due to the high cost of road transportation and large carbon emissions. Cold chain multimodal transport is a safe, efficient, and environmentally friendly form of cold chain transport organization, which can take into account transport cost, transport efficiency and carbon emissions. The cold chain of container multimodal transport development is rapid, especially the application of cold chain container multimodal transport in international trade more widely, but also the cold chain in improving the efficiency of cold chain transportation of container intermodal transportation. At the same time, many problems remain, such as the speed of various modes of transportation being easily affected by the external environment, and the lack of cohesion between various modes of transportation. However, carbon emissions from the comprehensive transportation system are a major part of the carbon emissions of the whole society. According to relevant statistics, in the carbon emissions from transportation, traffic congestion leads to the increase of energy consumption, resulting in additional carbon

emissions. The increase in carbon emissions due to road congestion is more than 20%. The prediction of carbon emission in a time-varying transportation network is uncertain due to the influence of many factors. Based on this, in the time-varying transportation network, considering the uncertainty of carbon emissions in the comprehensive transportation system, the cold chain container multimodal transportation route optimization model with carbon emissions in the time-varying network is constructed to reasonably deal with the existing problems of cold chain multimodal transportation under the time-varying network, so as to reduce the transportation cost of cold chain multimodal transportation and improve the efficiency of cold chain transportation. It plays an important role in optimizing cold chain multimodal transport network.

There are many research achievements on multimodal transportation route optimization worldwide. Tang, Y.Y. et al. [1] constructed a multi-objective 0-1 integer optimization model based on a multimodal transportation network and considering the characteristics of different transportation modes, taking transportation cost and transportation time as objective functions. Nsga-ii's algorithm was used to solve the model and the Pareto non-inferior solution set of the transport route was obtained. Zhang, X.L. et al. [2] established a multi-objective optimization model of multimodal transport route using a genetic algorithm taking transportation cost, time-value cost, and environmental cost as objective functions and hard and soft time Windows as constraints. Li, M. [3], considering the development status of multimodal transport in the Yangtze River Economic Belt, constructed a route optimization model for container multimodal transport in the Yangtze River Economic Belt in view of the low multimodal transport volume, poor connectivity of facilities and equipment, and insufficient shipping channel capacity. Hu, Y. et al. [4] looking at the problem of high logistics cost in long-distance commercial vehicle transportation, selected the appropriate mode of transportation from route optimization and determined the multimodal transportation route under the time constraint based on the principle of cost minimization. Liu, Q. et al. [5] constructed a multi-objective optimization model of a container multimodal transportation route along the Trunk Line of the Yangtze River and solved the model with a genetic algorithm. Li, M.L. et al. [6] established a multimodal transport allocation strategy model of emergency supplies based on the robust optimization theory, taking the small total cost of emergency rescue, response time, and penalty cost as the objective function. Li, J. et al. [7] comprehensively considered uncertain factors such as transportation time and customer demand and constructed an optimization model of multimodal transportation route under mixed uncertainty based on stochastic optimization theory. Considering carbon emission limitation, Liu, S. et al. [8] established a multimodal transportation route optimization model with the minimum total cost as the objective function, focusing on the limitation of departure times of railway and waterway transportation in the multimodal transportation network. A genetic algorithm was used to solve the model. Liu, D. et al. [9] constructed a multi-objective 0-1 programming model with a minimum total transportation cost, a minimum time cost and a minimum accident cost, and solved the model with a genetic algorithm. Wan, J. et al. [10] constructed a mixed integer multi-objective optimization model with transportation cost, transportation time, and logistics service quality as objective functions, and solved the model with a hybrid algorithm combining a genetic algorithm and an ant colony algorithm. Mnif et al. [11] used a fireworks algorithm to solve multi-objective multimodal transport problems, and compared and analyzed the solution results with CPLEX, concluding that the heuristic algorithm was more efficient. Chen, M.F. et al. [12] established the multimodal transportation network optimization model without considering safety stock and the multimodal transportation network optimization model considering safety stock, and updated the transportation price and demand information of the multimodal transportation network. Lozano et al. [13] constructed the shortest feasible route model of multimodal transport and solved the model through a sequential algorithm. Chen, Y. et al. [14] established the multimodal transport optimization model considering the actual constraints, and verified the model through an example. Assadipour et al. [15] used a heuristic algorithm to solve

multimodal transport route optimization model. Verga J. [16] considered the route blocking of multimodal transportation, constructed a multimodal transportation route optimization model, and adopted a heuristic algorithm to solve the model. Wang et al. [17] proposed a standardized calculation method for multimodal transport congestion, carbon emissions, and other problems, constructed a multimodal transport route optimization model, and provided a reasonable layout of a multimodal transport network and transport policy basis for low-carbon decision-making by evaluating the performance of transport nodes.

To sum up, this study develops a route optimization model of cold chain container multimodal transportation under a time-varying network on the basis of existing literature. The contribution of this study is mainly related to the following three aspects:

(1) The influence of time-varying factors on the traveling speed of vehicles in the transportation network is considered in the route selection of a cold chain container multimodal transportation, and the traveling speed of cars, trains and ships is analyzed.
(2) As the reality of the multimodal transport network is often not fully connected, a traditional genetic algorithm in the process of iteration produces many feasible solutions and the algorithm easily falls into local optimum. By inserting an heuristic algorithm to construct the initial population, elite reserved strategy and queen bee evolution crossover strategy are improved to make the traditional genetic algorithm more effective.
(3) Two-stage strategies of static and dynamic optimization are designed to solve the model, and in the dynamic environment, considering the impact of the time-varying traffic network on the comprehensive transportation system, the uncertainty of carbon emissions is taken into account. The carbon emissions of vehicles are calculated through the actual driving time.

The rest of this paper is organized as follows. In Section 2, we present the analysis and hypothesis of the route optimization problem of cold chain container multimodal transport under time-varying network. This section also introduces the travel time analysis of cold chain tools. Taking the lowest total cost as the objective function, an optimization model of cold chain container multimodal transportation route under time-varying traffic network was established. In Section 3, we design an improved algorithm solution model based on optimal preservation strategy and migration strategy. Section 4 designs the cold chain container multimodal transport in the Yangtze River Delta economic zone as an empirical analysis, uses the improved algorithm to solve the case in two stages of static and dynamic optimization, and discusses the calculation results. Section 5 concludes the paper.

2. Description and Modeling

2.1. Problem Description

We suppose that the multimodal transport network is composed of $G = \{I, L, J\}$, where I is the point set representing the nodes through which goods are transported, $i \in I$; L is the arc set, where $L = \{l_{i,i+1} | \forall i \in I\}$ and J are the sets of transport modes, where $j \in J, k \in J$ and O are the starting point of multimodal transport; D is the end point of multimodal transport,$O \in I, D \in I$. A regional intermodal network has n transport nodes with m modes of transport between each node. At the starting point O, *teu* reefer containers have to be transported to the end point D within the time window specified by the customer. The combination of transportation modes and transportation routes with transportation cost, carbon emission cost, a time window penalty cost, and the sum of damage cost of cold chain goods as the objective function can minimize the total cost of the transportation process. The transportation network diagram is shown in Figure 1.

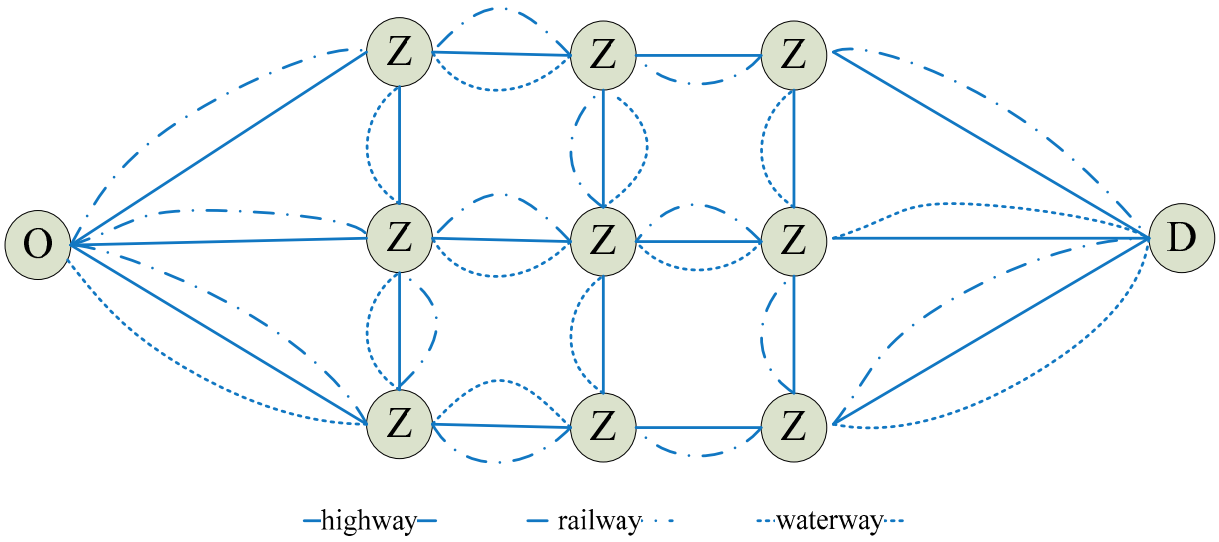


Figure 1. Transportation network diagram.

In this paper, the route optimization model of cold chain container multimodal transportation is established under a time-varying network. The actual process of multimodal transportation is complex. To simplify the complex model and enhance the operability of the model, the basic assumptions of the model are as follows:

(1) Only one transportation mode is selected between two nodes on the route of cold chain container multimodal transportation network.
(2) Cold chain containers can only be transferred at nodes, and one node can only be transferred once, and the same mode of transport cannot be transferred between.
(3) When the mode of transport changes at a node, the starting point of the new mode of transport and the end point of the original mode of transport are the same node.
(4) If there are m modes of transport between two nodes, there are m corresponding transport routes.
(5) The influence of the external environment on the speed of transport vehicles and the influence on the choice of multimodal transport routes are considered.
(6) Each mode of transportation of each node meets the volume of transportation, regardless of the limitation of transportation capacity.

2.2. Transport Time Analysis

2.2.1. Vehicle Speed Analysis

This study assumes that a shortest route exists between two nodes in the highway traffic network, the road congestion coefficient β_1 affects the vehicle speed V_i, and the route length D_{ij} and vehicle speed V_i affect the travel time $t_{(y)}$. Due to the time-varying characteristics of the highway traffic network, an unexpected traffic event occurs at point A when the vehicle travels on the shortest route i between two nodes to point T, as shown in Figure 2. The capacity of section TA is bound to decline, resulting in the vehicle travel time $t_{(y)}$ changing to $t_{(s)}$, and the travel time function of the shortest route i is $t_{(y)} = t_{(y)}OT + t_{(y)}TA + t_{(y)}AD = t_{(y)}OT + t_{(s)}TA + t_{(y)}AD$. If similar situations occur on multiple sections of the shortest route between two nodes, then $t_{(y)} = \sum t_{(y)} + \sum t_{(s)}$. $t_{(y)}$ is the driving time of vehicles on road section r without congestion, then $t_{(y)} = {}^{L_{ir}}/_{V_{ir}}$, where L_{ir} represents the length of road section r without congestion on the shortest route i, $\forall i; r$, and V_{ir} represents the actual driving speed of vehicles on road section r without congestion on the shortest route i, $\forall i; r$.

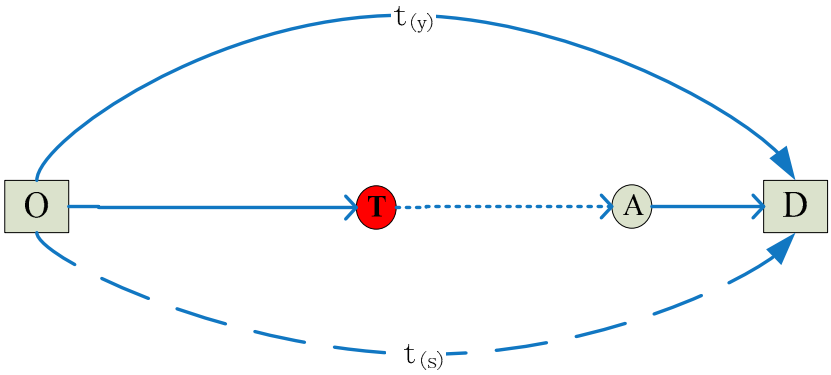


Figure 2. Schematic of emergencies encountered during vehicle driving.

It is assumed that the vehicle can reach the normal driving speed V_i on the road without congestion, and the steering coefficient μ is used to measure the driving difficulty on the actual road. Then, the vehicle driving speed V_{ir} is related to the times of turning left and right and passing the intersection.

$$\mu = (3\mu_L + \mu_R + 2\mu_C)/L_{ir} \tag{1}$$

where μ represents the steering coefficient, μ_C is the number of times through the intersection, μ_L is the number of left turns, and μ_R is the number of right turns.

$$V_{ir} = V_i \times (1 - 0.1\mu) \tag{2}$$

Based on the assumption that the speed of vehicles in the congested section m is V_{im}, the speed of vehicles V_{im} is related to the times of crossing the intersection and the congestion factor β.

$$V_{im} = V_i \times (1 - 0.1\mu) \times \beta_1 \tag{3}$$

This study assumes that road transport congestion coefficient β_1 is related to accident θ_{1h}, traffic jam θ_{2h}, weather θ_{3h}, vehicle performance θ_{4h}, and other emergencies θ_{5h}. Then, $\beta_1 = \theta_{1h} \times \theta_{2h} \times \theta_{3h} \times \theta_{4h} \times \theta_{5h}$.

2.2.2. Train Speed Analysis

The train speed is divided into three categories, namely, running, technical, and travel speed. The train running speed is the average speed of the train in the section, which is calculated by dividing the full distance of the railway between two stations by the travel time between two stations. The running time of the train between the two stations does not include the stopping time in the middle station and the additional time of starting and stopping. The calculation formula of the train running speed is as follows:

$$v_{run} = \frac{\sum nL}{\sum nt_{chunrun}} \tag{4}$$

Train travel speed is the average speed at which a train travels in one direction between two stations, which is calculated by dividing the total distance of the railway between the two stations by the time it takes to travel between them. The running time of a train between the two stations includes the stopping time in the middle station and the additional time of starting and stopping. The calculation formula of train travel speed is as follows:

$$v_{travel} = \frac{\sum nL}{\sum nt_{chunrun} + \sum nt_{qiting} + \sum nt_{zhongting}} \tag{5}$$

where $\sum nL$ represents the number of kilometers that the train runs in a certain section every day and night, $\sum nt_{chunrun}$ is the pure operating hours of the train in a certain section

every day and night, $\sum nt_{qiting}$ is the number of hours that the train starts and stops in a certain section every day and night, and $\sum nt_{zhongting}$ is the number hours that the train stops in a certain section every day and night.

$$v_{travel} = \beta_2 v_{run} \text{km/h} \tag{6}$$

In this study, it is assumed that the main factors affecting the train travel speed coefficient β_2 are the following: number of trains running ϑ_{1h}, number of stops ϑ_{2h}, number of overcrossing ϑ_{3h}, and distribution density of the boundary points for the rendezvous or overcrossing of trains ϑ_{4h}. Then, $\beta_2 = \vartheta_{1h} \times \vartheta_{2h} \times \vartheta_{3h} \times \vartheta_{4h}$.

2.2.3. Ship Speed Analysis

Based on the assumption that the normal speed of a ship on the shortest route is $v_{(y)ship}$, and accidents and other related factors affect the normal speed, the speed of the sheep is $v_{(s)ship}$, where

$$v_{(s)ship} = v_{(y)ship} \times \beta_3 \tag{7}$$

In this study, we assume that the influence coefficient of the ship's average speed β_3 is related to accident ρ_{1h}, natural disaster ρ_{2h}, ship's performance ρ_{3h}, full capacity ρ_{4h}, and other factors (wind speed, wind direction, velocity, and flow direction)ρ_{5h}. Then, $\beta_3 = \rho_{1h} \times \rho_{2h} \times \rho_{3h} \times \rho_{4h} \times \rho_{5h}$.

2.3. Model Symbols and Parameters

Symbol descriptions of the multi-objective optimization model of the railway cold chain transportation route based on dynamic freight train information are presented in Table 1.

Table 1. Notations for the model.

Notation	Definition
$x_{i,i+1}^{j}$	The value is 1 if the j mode of transport is used from node i to node $i+1$, and 0 otherwise
$l_{i,i+1}^{j}$	Cold chain container from node i to node $i+1$ using the j mode of transport distance
$t_{i,i+1}^{j}$	Cold chain container from node i to node $i+1$ using the j mode of transport time
$e_{i,i+1}^{j}$	Cold chain container from node i to node $i+1$ using the j mode of unit transport cost
$co_{i,i+1}^{j}$	Cold chain container from node i to node $i+1$ using the j mode of unit time carbon emission
co_z	Carbon emissions of cold chain container loading and unloading equipment during node transfer
$y_i^{j,k}$	This value is 1 if we switch from mode j to mode k at i, and 0 otherwise
$e_i^{j,k}$	If the mode of transport is changed from j to k at i, the cost per unit of cargo is 0 if the mode of transport is not changed at that point
$t_i^{j,k}$	If the mode of transport is changed from j to k at i, the reloading time per unit of cargo is 0 if the mode of transport is not changed at that point
e_l	Unit cost of refrigerating a reefer container
co_l	Carbon emissions per unit of refrigerated container cooling

Table 1. *Cont.*

Notation	Definition
r_e	A carbon tax value
γ_1	Unit storage cost for reefer containers arriving early at i node
γ_2	Unit penalty charge for late arrival of reefer container to i node
a_i	Earliest time allowed to reach i node
b_i	Latest time allowed to reach i node
EE	Lower limit of time from start O to end D for reefer containers
EL	Upper limit of time from start O to end D for reefer containers

2.4. Model Building

Based on the preceding assumptions and combined with the actual operation process of cold chain container multimodal transport, the objective function of the model is as follows:

(1) Considering the total cost of the transportation process, including the transportation cost of cold chain containers between nodes, the renodement cost of reefer containers at nodes, and the refrigeration cost of reefer containers. The minimum transportation cost C_1 is

$$C_1 = \sum_{i=1}^{n}\sum_{j=1}^{m} e_{i,i+1}^{j} x_{i,i+1}^{j} l_{i,i+1}^{j} + \sum_{i=1}^{n}\sum_{j=1}^{m}\sum_{k=1}^{m} y_i^{j,k} e_i^{j,k} + e_l\left(\sum_{i=1}^{n}\sum_{j=1}^{m} t_{i,i+1}^{j} x_{i,i+1}^{j} + \sum_{i=1}^{n}\sum_{j=1}^{m}\sum_{k=1}^{m} y_i^{j,k} t_i^{j,k}\right) \tag{8}$$

(2) Carbon emission cost in the transportation process, including the carbon emission cost of cold chain container transportation between nodes, reefer container reloading at nodes, and reefer container refrigeration. Among them, in the time-varying network, the travelling speed of transport vehicle is affected by the change of time-varying factors, leading to the delay of travelling time, which leads to the uncertainty of the carbon emissions of the transport vehicle. Therefore, this paper calculates the carbon emission through the actual travelling time of transport vehicle, and converts the distance between nodes into time through $l_{i,i+1}^{j}/v_{(s)i,i+1}^{j}$, where $v_{(s)i,i+1}^{j}$ is the actual traveling speed of the cold chain container when j transportation mode is adopted from node i to node $i+1$ in the time-varying network. Then, the minimum carbon emission cost C_2 is

$$C_2 = \left(\sum_{i=1}^{n}\sum_{j=1}^{m} co_{i,i+1}^{j} x_{i,i+1}^{j} l_{i,i+1}^{j} + \sum_{i=1}^{n}\sum_{j=1}^{m}\sum_{k=1}^{m} y_i^{j,k} co_z + co_l\left(\sum_{i=1}^{n}\sum_{j=1}^{m} t_{i,i+1}^{j} x_{i,i+1}^{j} + \sum_{i=1}^{n}\sum_{j=1}^{m}\sum_{k=1}^{m} y_i^{j,k} t_i^{j,k}\right)\right) r_e \tag{9}$$

(3) Time window cost, including unit storage fee for the reefer container arriving at the transfer node in advance and the unit penalty fee for the reefer container being late, then the minimum time window cost C_3 is

$$C_3 = \gamma_1 \sum_{i=1}^{n} \max(a_i - t_i, 0) + \gamma_2 \sum_{i=1}^{n} \max(t_i - b_i, 0) \tag{10}$$

(4) Cost of damage. Cold chain goods storage conditions are strict, sensitive to environmental changes; time and temperature changes have a great impact on the deterioration of goods. The metamorphic cost is positively correlated with both changes. In the process of cold chain container multimodal transportation, the external environment is the main link that causes the temperature change in the container and leads to the deterioration of goods. In this paper, the Arrhenius formula is used to calculate the deterioration cost of fresh goods due to temperature change [18]. The formula is: $K = Ae^{-\left(\frac{E_a}{RT}\right)}$, where K is the reaction rate constant that changes with temperature,

and A is the frequency factor; Ea is the reaction activation energy, T is the absolute temperature, and R is the molar gas constant.

It is assumed that under the condition of constant temperature, the deterioration rate of fresh perishable products remains constant, but it will change exponentially with time accumulation (Shi S. [19]). Therefore, its metamorphic function is as follows:

$$G_{teu}(t) = G_{teu}(0) \times K \times e^{-ut_z} \tag{11}$$

where $G_{teu}(t)$ represents the mass of the reefer container at time t, and $G_{teu}(0)$ is the mass of the reefer container at time 0; that is, when the product is intact. t_z is the total transportation time of the cold chain container, K is the reaction rate constant that changes with the temperature, and μ is the sensitivity coefficient, that is, the sensitivity of the cold chain goods to time.

$$\mu = K_{\max} \exp(-\frac{E_a}{RT}) \tag{12}$$

In the preceding metamorphic function, the more sensitive the product is to time, the smaller the sensitivity coefficient is. Otherwise, the larger the sensitivity coefficient. Therefore, the cost of cargo damage in the multimodal transport of cold chain containers C_4 is expressed as:

$$C_4 = P_{teu}[1 - e^{-\mu \frac{l^j_{i,i+1}}{v_j}}] \tag{13}$$

To sum up, the cost model of cold chain container multimodal transportation under time-varying network is as follows.

$$\min C = C_1 + C_2 + C_3 + C_4 \tag{14}$$

s.t.

$$t^j_{(y)i,i+1} = \frac{l^j_{i,i+1}}{v_j}, v_j \in (v_{(y)highway}, v_{(y)railway}, v_{(y)waterway}) \quad , \forall i \in I, \forall j \in J \tag{15}$$

$$t^j_{(s)i,i+1} = \frac{l^j_{i,i+1}}{v_j}, v_j \in (v_{(s)highway}, v_{railway}, v_{(s)waterway}) \quad , \forall i \in I, \forall j \in J \tag{16}$$

$$x^j_{i,i+1} + x^j_{i-1,i} \geq 2y^{j,k}_i \quad \forall i \in I, \forall j,k \in J \tag{17}$$

$$t^j_{i,i+1} \geq 0, t^{j,k}_i \geq 0, l^j_{i,i+1} \geq 0 \tag{18}$$

$$\sum_{j=1}^{m} x^j_{i,i+1} = 1 \quad \forall i \in I, \forall j \in J \tag{19}$$

$$\sum_{j=1}^{m} \sum_{k=1}^{m} y^{j,k}_i = 1 \quad \forall i \in I, \forall j,k \in J \tag{20}$$

$$x^j_{i,i+1}(t_{i+1} - t_i) \geq 0 \quad \forall i \in I \tag{21}$$

$$EE \leq \sum_{i=1}^{n} \sum_{j=1}^{m} x^j_{i,i+1} t^j_{i,i+1} + \sum_{i=1}^{n} \sum_{j=1}^{m} \sum_{k=1}^{m} y^{j,z}_i t^{j,z}_i \leq EL \tag{22}$$

$$x^j_{i,i+1} = \begin{cases} 1, thej\ transport\ mode\ is\ selected\ from\ node\ i\ to\ node\ i+1 \\ 0, thej\ transport\ mode\ is\ not\ selected\ from\ node\ i\ to\ node\ i+1 \end{cases} \tag{23}$$

$$y^{j,k}_i = \begin{cases} 1, at\ node\ i, the\ transport\ mode\ is\ transformed\ from\ j\ to\ k \\ 0, \quad otherwise \end{cases} \tag{24}$$

Equation (14) expresses the minimum total cost in the transportation process, which is the objective function of the model. It mainly includes for these types of cost: transportation, carbon emission, time window penalty cost, and refrigerated goods damage cost. Equation (15) expresses the calculation of transport time from node i to node $i+1$ of the transport vehicle in a static environment, which is related to the traveling speed of the transport vehicle. Equation (16) expresses the calculation of transport time from node i to node $i+1$ of the transport vehicle in a dynamic environment, which is related to the traveling speed of the transport vehicle. Equation (17) shows the continuity and accuracy of cold chain container multimodal transportation. Equation (18) demonstrates that transport time, transfer station time, and transport distance are all positive. Equation (19) shows that only one transportation mode can be selected between two adjacent nodes of the cold chain container during transportation. Equation (20) shows that reefer containers can only be converted from one mode of transportation to another mode if they are transferred at a certain node. In Equation (21), to ensure the continuity of cold chain container transport in time, the last arrival time at node $i+1$ is later than the first arrival time at node i. Equation (22) refers to the total time (transportation time and waiting time for reefer containers) from the starting point O to the end point D, which should be within the time window required by the customer. Equation (23) is a decision variable that indicates whether a certain mode of transportation is adopted between two nodes. Equation (24) is a decision variable that indicates whether to change the mode of transportation at a node.

3. Algorithm Design

3.1. Calculation of Travel Time

In time-invariant networks and the symmetrical route model, as in section (b_i, b_{i+1}), the motion of the transport time is $t^j_{(y)i,i+1} = d^j_{(y)i,i+1}/\overline{v}^j_{(y)i,i+1}$, $\overline{v}^j_{(y)i,i+1}$ of section (b_i, b_{i+1}) of the means of transport is expected to average speeds, but in the time-dependent multimodal transport network model, accidents, traffic, weather conditions, or other emergencies such as time-varying factors affect the transportation time. Assuming that the departure time is t_0, the distance is $d^j_{i,i+1}$, and the traveling speed of the vehicle is similar to that in Figure 1, it is divided into multiple sections. If section TA is section k, then the actual traveling speed is $v^j_{(s)ki,i+1}$, and the corresponding departure time interval is $[t_k, t_{k+1}]$, the calculation process of arrival time and traveling time can be expressed as follows [8]:

(1) The value of k is determined according to $t_0 \in [t_k, t_{k+1}]$, the corresponding $v^j_{(s)ki,i+1}$ is obtained, the value of $d^j_{i,i+1} \to d$ is assigned, and $t' = t_0 + d/v^j_{(s)ki,i+1}$ is calculated.

(2) If $t' \le t_{k+1}$ is true, then the output arrival time t' and travel time $t' - t_0$ end the cycle. Otherwise, step (3) is undertaken.

(3) $d^j_{i,i+1} - v^j_{(s)ki,i+1}(t_{k+1} - t_k) \to d$ is updated, $t' = t_{k+1} + d/v^j_{(s)(k+1)i,i+1}$ is calculated, $k+1 \to k$ is calculated, and step (2)) is revisited.

3.2. Algorithm Design

Considering that the genetic algorithm is a random search algorithm that relies on an initial solution, easily falls into local optimum, and has slow convergence speed when solving large-scale complex optimization problems, this study designed an insertion heuristic algorithm to construct the initial population, and improve the elite retention strategy and queen evolution crossover strategy. Specifically, the algorithm can be described as follows. On the one hand, to reduce the dependence of the genetic algorithm on the initial solution, in the process of constructing the initial solution, this study adopts the insertion heuristic algorithm to construct the initial population and improve the excellence of the initial solution. On the other hand, to reduce the loss of highly adaptable offspring in the crossover and mutation process of the algorithm, the queen evolution operation and crossover operation of the genetic algorithm complement each other's advantages and

improve the elite retention strategy to improve the optimization ability and convergence speed of the algorithm.

(1) Chromosome coding. In this study, a multi-parameter joint coding method is adopted. In the coding process, a chromosome is divided into two parts. The first part is route coding, which indicates whether all cities have gone through 0–1 coding (the first and last nodes are determined to be 1, the middle nodes that have not gone through are coded to be 0, and the passed nodes are coded to be 1). The second part is generated according to the first part. The number of generated nodes is the number of urban nodes minus 1, indicating the mode of transport (expressed by 1–3). No air transport mode code exists, and 1–3 represent the three modes of transport, namely, highway, railway, and water transport. The corresponding chromosomes in the first and second parts are shown in Figure 3. The two corresponding chromosomes in the first part 1-1-0-0-1-0-1-1-0-0-1 and the second part 1-2-3-1-2 are shown as follows: Road transport is selected from node 1 to node 2, railway transport is selected from node 2 to node 5, water transport is selected from node 5 to node 7, road transport is selected from node 7 to node 8, and railway transport is selected from node 8 to node 11. So, in this pattern, the mode of transport part codes the mode of transport on the line. In this way, any multimodal transport scheme from the beginning to the end can be represented.

(2) Initialization of the population. The randomly generated individual consists of two parts, among which the first part is the transportation line code, the starting point of transportation is set as 1, and the remaining nodes are numbered as 2,3 in sequence, n. The second part is transportation mode coding, randomly generating gene values at each gene location (value range: 1-highway, 2-railway, and 3-waterway). We judge whether the initial individual is feasible, and the initial group is the individual with no infinite transport distance in the corresponding route of the current individual.

(3) Fitness calculation. The two parts of each chromosome are decoded and converted into a matrix (node 1, node 2, mode of transport) to calculate the total cost under different modes of transport. For the constraints of the transportation time limit and maximum transportation cost, the transportation time and transportation cost are, respectively, multiplied by the penalty coefficient 1.5 according to the route that does not meet the constraint conditions obtained by the decoding route. Otherwise, the penalty coefficient is unchanged. In the case of the transfer times constraint, the number of transfer times is counted and the penalty cost is added to the total cost if the number of transfer times exceeds the constraint. In this study, the transit times are set as 5. When the route transit times exceed 5, a penalty value of positive infinity is added to the objective function. The objective function of the feasible solution of the worst individual in the c-generation population is M_c, and the fitness function of any individual i of the current population is set as shown in Equation (25).

$$fitness_c^i = \begin{cases} 1/M_c^i, M_c^i \le Z \\ 1/(\lambda M_c^i), M_c^i \ge Z \end{cases} \quad (25)$$

where Z is a positive number large enough, λ is the control parameter, and $\lambda > 1$. The fitness function tries to put the high-quality individuals from feasible solutions into the new population, but the individuals from non-feasible solutions may also be put into the new population. The higher the value of λ is, the less possibility the non-feasible solutions will enter the new population.

(4) Select operation. To ensure the proportion of feasible solutions in the new population and the diversity of feasible solutions in the population, this study improves the genetic algorithm based on the optimal preservation strategy. At the end of each iteration, the best individuals in the current offspring population were saved into the elite database so that the excellent individuals in the iteration process could be preserved. Before each iteration, the parent population was fused with some

individuals in the elite pool to make the population evolve in a better direction. To ensure the optimality of the elite library, the threshold NUM of the elite library is set to ensure its optimality and the convergence speed of the algorithm. The roulette method is introduced to screen the selected individuals from the elite pool each time, which further speeds up the convergence of the algorithm.

(5) Crossover operation. To further improve the optimization capability of the algorithm, the optimal individuals in the iterative process were crossed based on the crossover operation of queen bee evolution, so that the excellent gene fragments could be preserved. Before the crossover operation in each iteration, the individual with the highest fitness value was selected as the queen bee, and the GF_{NUM} individual was selected as the worker bee from the parent population by the roulette method, and two chromosome fragments were randomly selected for crossover and interchange operation. The transit and non-transit nodes were swapped, and the transport mode constraints, code quantity constraints, and capacity constraints were checked. If the constraints were not met, gene fragment repair was carried out to make the crossover individuals meet all constraints [20]. Therefore, the schematic of the queen evolution crossover operation designed in this study is shown in Figure 4.

(6) Mutation operation. The function of the mutation operator is to improve the change driving mechanism of population evolutionary diversity. Based on the problem of cold chain multimodal transport under time-varying network conditions considered in this study, some segments of crossed individuals were selected for the mutation operation according to a certain mutation probability P_M. First, a fragment of a chromosome is selected and reversed. After the reverse operation, the transfer nodes of multimodal transport in chromosome fragments were extracted. Finally, based on the constraints of transport mode and transport times in a multimodal transport network, a new chromosome was obtained by combining a single-point random variation and a two-point exchange variation.

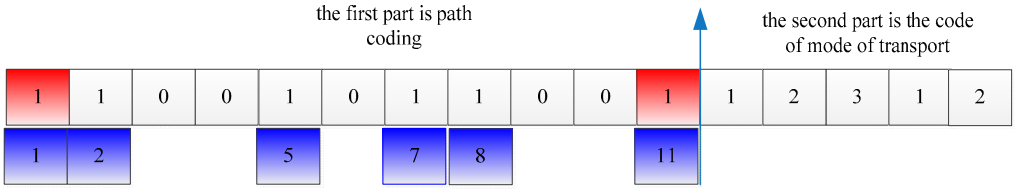


Figure 3. Chromosome coding diagram.

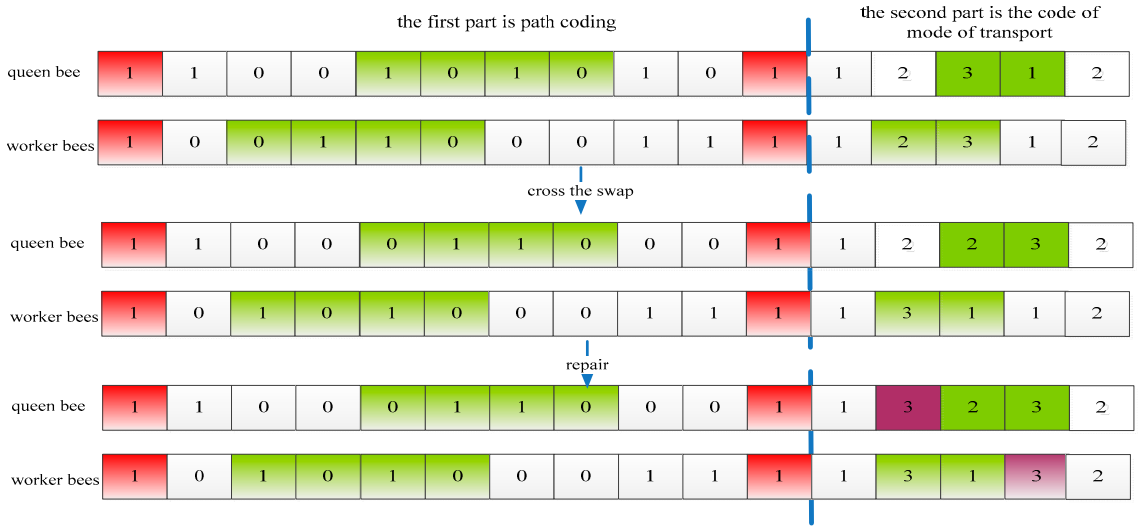


Figure 4. Diagram of the queen bee evolution.

Based on the preceding description, the specific steps of the improved genetic algorithm designed in this study are shown as follows:

① The following parameters are set: population size MAX_{pop}, maximum iteration number C_{max}, crossover probability P_C, mutation probability P_M, and elite library capacity NUM.

② The insertion heuristic algorithm is used to generate the initial population, perform the initial screening and decoding for each individual with capacity constraints, calculate the fitness value, and initialize the iteration number C = 0.

③ The parent population was sorted according to the fitness value. The individuals with the highest fitness value were stored in the elite database, and some individuals were selected from the parent population for crossover and mutation operation according to the roulette method.

④ The individual with the maximum fitness value in the current parent population was selected as the queen, and $P_C * MAX_{pop}$ individuals were selected to form the worker colony by roulette. Each worker bee was cross-operated with the queen, and $2 * P_C * MAX_{pop}$ offspring were generated after cross-operation.

⑤ Some individuals from the offspring population were selected after crossover according to the mutation probability for mutation operation, and NUM individuals were selected from the elite pool by roulette.

⑥ The fitness value of individuals after mutation was calculated and sorted according to the fitness value from large to small. The former $MAX_{pop} - NUM$ individuals with the largest fitness value and NUM individuals selected from the elite database were selected to form the next-generation parent population to update the elite database, and the iteration times were $C = C + 1$.

⑦ If the iteration number C is greater than the maximum iteration number C_{max}, the iteration is stopped. Otherwise, proceed to ③.

4. Empirical Analysis

4.1. Data Description

In this study, 27 cities (with an area of 225,000 sq. km.) in the Yangtze River Delta urban agglomeration were taken as the research object. To verify the feasibility of the model and algorithm, a multimodal transport network of 27 node cities in the Yangtze River Delta urban agglomeration is constructed, and its transport route is shown in Figure 5.

Table 2. Summary of transport distance of each section.

9	Terminal	Highway	Railways	Waterway	Starting Point	Terminal	Highway	Railways	Waterway
Shanghai	Nantong	126	126	128	Huzhou	Jiaxing	78.1	78.1	—
Shanghai	Suzhou	102	102	170	Huzhou	Hangzhou	85.1	85.1	—
Shanghai	Jiaxing	98	98	80	Hangzhou	Jiaxing	83.6	83.6	—
Shanghai	Zhoushan	—	—	40	Hangzhou	Shaoxing	49.2	49.2	—
Nantong	Yancheng	187.5	187.5	191	Hangzhou	Jinhua	162.9	162.9	—
Yancheng	Taizhou	126.2	126.2	—	Jinhua	Shaoxing	159.3	159.3	—
Yancheng	Yangzhou	167.6	167.6	—	Jinhua	Taizhou	213.3	213.3	—
Yangzhou	Taizhou	71.8	71.8	—	Jinhua	Wenzhou	214.5	214.5	—
Yangzhou	Zhenjiang	36	36	—	Wenzhou	Taizhou	215	215	210
Yangzhou	Nanjing	84	84	—	Taizhou	Shaoxing	218.4	218.4	—
Yangzhou	Changzhou	98.8	98.8	—	Taizhou	Ningbo	168.3	168.3	180
Taizhou	Wuxi	128.1	128.1	—	Ningbo	Jiaxing	148.6	148.6	—

Table 2. *Cont.*

9	Terminal	Highway	Railways	Waterway	Starting Point	Terminal	Highway	Railways	Waterway
Taizhou	Suzhou	158.1	158.1	—	Ningbo	Hangzhou	150.4	150.4	—
Suzhou	Jiaxing	82.4	82.4	—	Ningbo	Zhoushan	82.4	82.4	78
Suzhou	Huzhou	85	85	80	Nanjing	Zhenjiang	65.6	65.6	87
Suzhou	Wuxi	39.3	39.3	18	Nanjing	Ma'anshan	56.9	56.9	48
Wuxi	Huzhou	144	144	121	NanJing	Chuzhou	58.9	58.9	—
Wuxi	Changzhou	62.3	62.3	77	Chuzhou	Hefei	128.4	128.4	—
Changzhou	Zhenjiang	79	79	58	Hefei	Ma'anshan	149.3	149.3	—
Changzhou	Xuancheng	194	194	—	Hefei	Anqing	162.5	162.5	—
Changzhou	Huzhou	139.8	139.8	—	Anqing	Chizhou	66.3	66.3	60
Wuhu	Zhenjiang	154.1	154.1	—	Chizhou	Tongling	50.9	50.9	36
Wuhu	Xuancheng	63.1	63.1	—	Tongling	Wuhu	90.5	90.5	108
Xuancheng	Zhenjiang	175.5	175.5	—	Wuhu	Ma'anshan	42.4	42.4	48

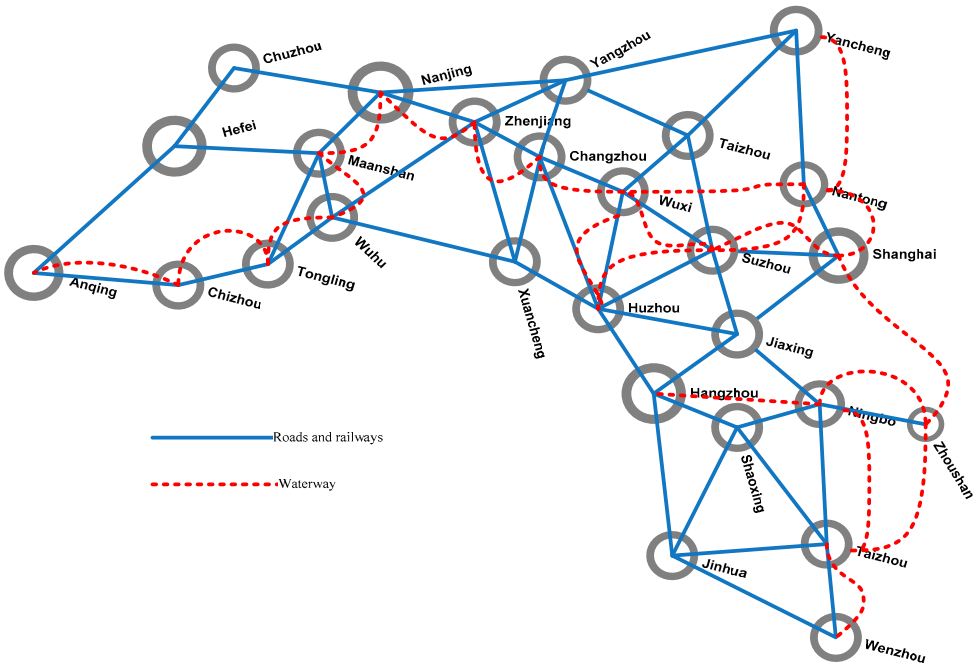


Figure 5. Multimodal transport network.

According to Amap, the China Railway Information Network and other relevant maps, the shortest path distance between the 27 nodes of the Yangtze River Delta urban agglomeration in road, railway and waterway mode is obtained, as shown in Table 2.

A transportation enterprise undertakes three transportation tasks as shown in Table 3. It is known that the weight of goods per TEU is 20 t and the total value of goods is 150,000 yuan. In transportation task 1, the starting point of transportation is Shanghai and the end point is Nanjing. The other 25 node cities are selected as possible transit nodes. For convenience of expression, these 25 cities are numbered as 2, 3, . . . , 26. Transport task 2: Transport starts from Shanghai City and ends at Hefei City; Transport Task 3: Transport starts from Shanghai City and ends at Jinhua City. The coding principle of transit nodes is the same as that of transport Task 1.

Table 3. Transportation task list.

Transportation Task	Starting Point	End Node	Number	Value of Goods (Yuan)
Task I	Shanghai	Nanjing	1TEU	150,000
Task II	Shanghai	Hefei	1TEU	150,000
Task III	Shanghai	Jinhua	1TEU	150,000

We referred to the freight rate table of freight service, the IPCC National Greenhouse Gas Inventory Guide, the China Transport Yearbook, and other data to determine the speed, unit transport cost and carbon emission of road, rail, and water transport. The unit freight, transport speed and unit carbon emission of the three modes of transport are shown in Table 4.

Table 4. Overview of three modes of transport.

Mode of Transportation	Highway	Railway	Waterway
Unit freight (yuan/teu × km)	8.5	2.025	0.5
Transportation speed (km/h)	80	60	30
Carbon emissions (kg/teu × km)	1.4	0.412	0.332

By sorting out and analyzing the relevant literature and data, as well as referring to the specific situation of ports and transit stations, the corresponding transport parameters between highway, railway, and waterway transport modes are obtained, including transport cost, transport time, and transport carbon emissions. The transport costs, carbon emissions, and transport time among the three modes of transport are shown in Table 5.

Table 5. Reloading between different modes of transport.

Mode of Transportation	Highway			Railway			Waterway		
	Cost (Yuan)	Carbon (kg)	TIME (h)	Cost (Yuan)	Carbon (kg)	Time (h)	Cost (Yuan)	Carbon (kg)	Time (h)
Highway	0	0	0	30	1.4	0.1	50	3.5	0.3
Railway	30	1.4	0.1	0	0	0	60	4.2	0.4
Waterway	50	3.5	0.3	60	4.2	0.4	0	0	0

The algorithm parameters are set as follows: the crossover rate is 0.7, the mutation rate is 0.3, the population size is 100, and the maximum iteration time is 800. Matlab2014b software is used to solve static model and dynamic model, respectively.

4.2. Static Model Solution

A static model refers to the minimum optimization of the total cost of a multimodal transport carrier considering transport cost, carbon emission cost, time window cost, and cargo damage cost in the deterministic mode without considering the influence of traffic congestion, external environment, and transport vehicle failure on the transport time of various modes. Matlab2014b was used to solve the three tasks in 28.44 s. The optimal route of multimodal transportation is shown in Table 6. The minimum total cost of the three tasks is 13,286.03 yuan. The cost convergence curve of multimodal transport for three tasks in static mode is shown in Figure 5, where (a) is the convergence curve of task I, (b) is the convergence curve of task II, and (c) is the convergence curve of task III. The abscissa is the number of iterations, and the ordinate is the total transportation cost of the objective function (unit: yuan).

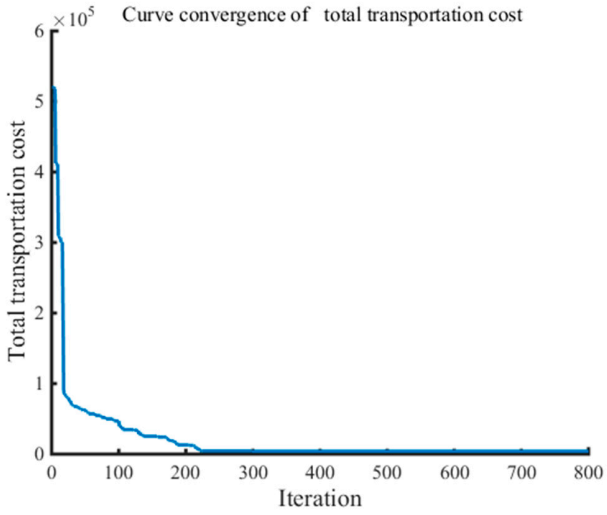
Table 6. Multimodal transport route with lowest total cost in the static model.

Transportation Task	Transport Route	Cost (Yuan)	Algorithm Time (S)
Task I	Shanghai→highway→Suzhou→highway→Wuxi →highway→Changzhou→highway→Zhenjiang →highway→Nanjing	4043.11	9.1
Task II	Shanghai→highway→Suzhou→highway→Wuxi →railway→Changzhou→railway→Zhenjiang →railway→Nanjing→railway→Tuzhou →railway→Hefei	5771.38	11.1
Task III	Shanghai→highway→Jiaxing→highway →Hangzhou→railway→Jinhua	3471.54	8.3

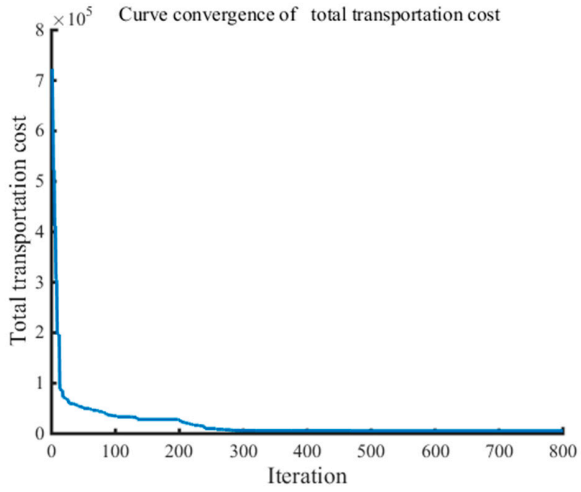
4.3. Dynamic Model Solution

When the time variable factors change, the optimization scheme in the static environment leads to the change of the optimal route of multimodal transport. At this time, target optimization should be carried out again according to the real-time information after the change of the multimodal transport network. The steering coefficient of highway driving in the Yangtze River Delta urban agglomeration on a certain day is assumed to be 0, the average congestion coefficient is 0.76, the accident coefficient is 0.85, the weather coefficient is 0.91, the vehicle performance coefficient is 0.99, and other emergency coefficients are 0.90. The coefficient of running volume of passenger and freight trains, the coefficient of stopping times, the coefficient of crossing times, and the coefficient of crossing times of passenger and freight trains affecting the speed of train travel are 0.95, 0.94, 0.94, and 0.99, respectively. Accident coefficient 1, natural disaster coefficient 1, ship performance coefficient 0.95, ship load factor 0.9, and other coefficients (wind speed, wind direction, velocity, flow direction, and others) are 0.9. Under the dynamic model, when the multimodal transport carrier considers the minimum optimization of the total cost of transportation cost, carbon emission cost, time window cost, and cargo damage cost, MATLAB2014B takes 30.9 s to solve the three tasks. The optimal route of multimodal transport is shown in Table 5, and the minimum total cost of the three tasks is 14,757.77 yuan. The convergence curve of the total cost of multimodal transportation of three tasks under the static model and dynamic model is shown in Figures 6 and 7, where (a) is the convergence curve of task I, (b) is the convergence curve of task II, and (c) is the convergence curve of task III. The abscissa is the number of iterations, and the ordinate is the total transportation cost of the objective function (unit: yuan).

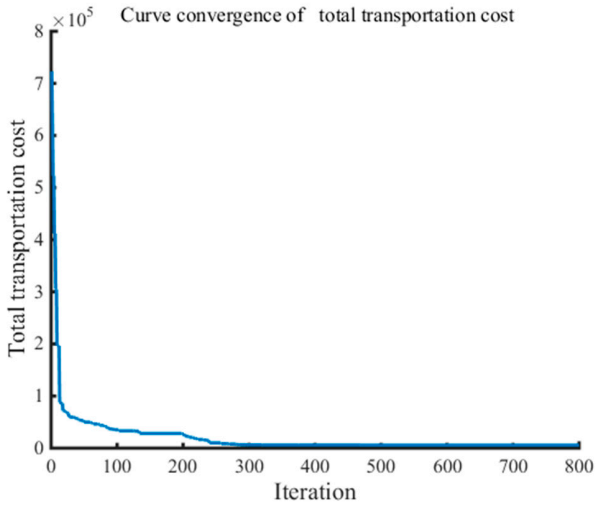
Table 6 shows that the total cost of the optimal route for cold chain multimodal transport in a static environment is 13,286.03 yuan. As can be seen from Table 7, the total cost of the optimal route of cold chain multimodal transportation in a dynamic environment is 14,757.77, which increases by 11.07% due to the influence of time-varying factors on the traveling speed of transport vehicles. The results show that road transportation has adverse effects on transportation costs and carbon emissions, but the disadvantages of road transportation become advantages when the penalty coefficient of time window cost is larger and the damage cost of the cold chain is considered. When the transport network is affected by the external environment, the road transport is more sensitive to the external environment, while the railway transport is relatively less affected by the external environment, which is conducive to the realization of the "turn-to-rail" transport mode.



(**a**)

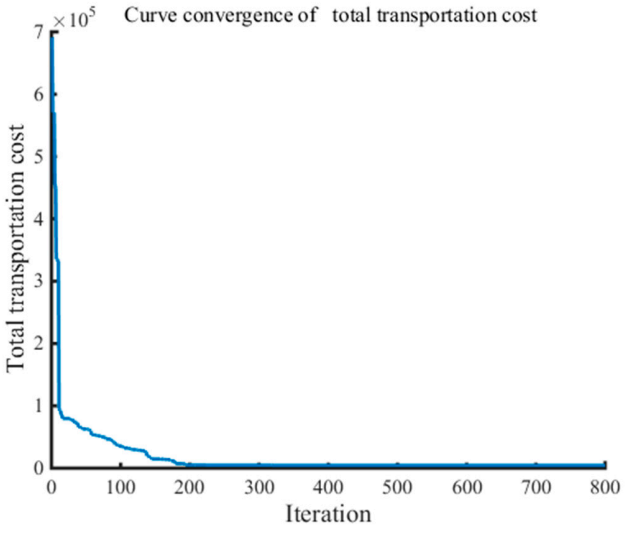


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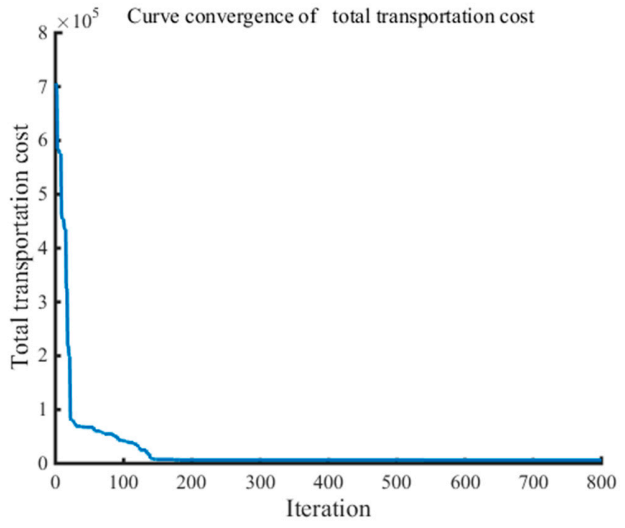


(**c**)

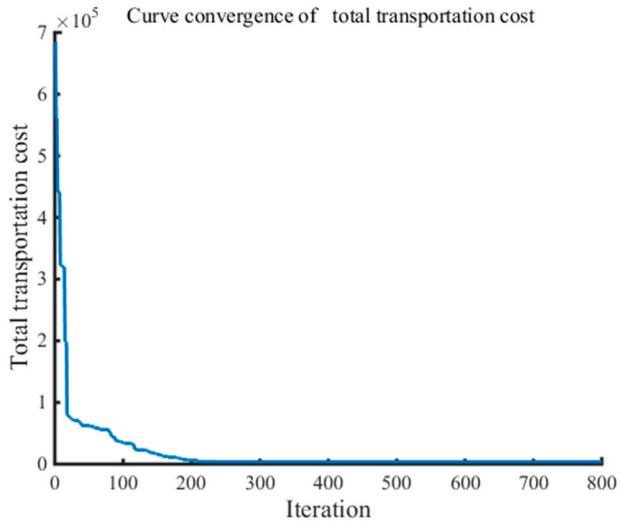
Figure 6. Convergence curves of multimodal transport costs for three tasks under the static model (Unit: Yuan). (**a**) Task I, (**b**) Task II, (**c**) Task III.



(**a**)



(**b**)



(**c**)

Figure 7. Convergence curves of multimodal transport costs for three tasks under dynamic model (Unit: Yuan). (**a**) Task I, (**b**) Task II, (**c**) Task III.

Table 7. Dynamic model of the lowest total cost of the multimodal transport route.

Transportation Task	Transport Route	Cost (Yuan)	Algorithm Time (s)
Task I	Shanghai→waterway→Nantong→waterway →Changzhou→waterway→Zhenjiang→waterway →Nanjing	4525.82	10.1
Task II	Shanghai→railway→Suzhou→railway→Wuxi →railway→Changzhou→railway→Zhenjiang →railway→Nanjing→railway→Tuzhou →railway→Hefei	6404.79	11.2
Task III	Shanghai→railway→Jiaxing→railway→Hangzhou →railway→Jinhua	3827.16	9.6

4.4. Analysis of Experimental Results

To verify the effectiveness of the algorithm, a catastrophic adaptive genetic algorithm (CAGA) and a genetic algorithm (GA) are written using Matlab R2014b, in which the fitness function is constructed based on the Monte Carlo sampling method; adaptive crossover and mutation operations are designed, and a catastrophic operator is added to prevent premature convergence and improve the global search performance of the algorithm. The experiment was compared with the queen bee genetic algorithm (QBGA) designed in this paper. Under the same experimental conditions, the transportation route, cost, and time of cold chain container multimodal transportation in static environment are calculated by the three methods, as shown in Table 8.

Table 8. Algorithm comparison.

Scenario	Algorithm	Scenario I: Static Environment			Scenario II: Dynamic Environment		
		Total Cost (Yuan)	Transit Times	Algorithm Time (s)	Total Cost (Yuan)	Transit Times	Algorithm Time (s)
Task I	QBGA	4043.11	0	9.1	4525.82	0	9.6
	GA	4798.65	1	10.3	5242.74	1	10.1
	CAGA	4048.33	0	10.5	4554.36	0	10.8
Task II	QBGA	5771.38	1	10.1	6404.79	0	10.2
	GA	6311.83	1	11.1	7176.22	1	11.2
	CAGA	5771.4	1	12.3	6420.49	0	12.2
Task III	QBGA	3471.54	1	8.3	3827.16	0	9.1
	GA	3806.64	1	9.4	4365.09	1	9.6
	CAGA	3806.64	1	10.4	4365.09	1	10

As shown in Table 7, based on two scenarios of static environment and dynamic environment, the advantages of the queen bee evolutionary genetic algorithm designed in this paper are shown in the following two aspects through analysis and comparison with the traditional genetic algorithm and the catastrophe adaptive genetic algorithm:

(1) Cost objective function value: Aiming at the static environment and dynamic environment, the objective function value obtained by using the queen evolutionary genetic algorithm, traditional genetic algorithm, and catastrophe adaptive genetic algorithm, respectively, show that queen evolutionary genetic algorithm has improved by 12.27 and 13.73% in terms of numerical value compared with the traditional genetic algorithm. The objective function value of the queen bee evolutionary genetic algorithm was similar to that of the catastrophe adaptive genetic algorithm, but the convergence speed of the queen Bee evolutionary genetic algorithm is improved by 17.28 and

12.48%. In addition, considering the influence of the time-varying traffic network on vehicle speed under the dynamic environment, the vehicle needs to choose a transportation mode with better cost according to road congestion, which puts forward higher requirements for the algorithm. Among them, the traditional genetic algorithm easily falls into the local optimal solution when facing this complex problem, which cannot meet the needs of solving. The queen bee evolutionary genetic algorithm has a strong search ability in the face of highway, railway, and water transportation in a complex time-varying traffic network and can better find out the cost of a multimodal transport route, thus can provide multimodal transport enterprises to choose scientific and reasonable multimodal transport route decision-making and implementation; authors provide a more comprehensive and effective decision-making efficiency.

(2) Convergence speed: The queen bee evolutionary genetic algorithm has a strong search ability, and can solve this kind of combinatorial optimization problem more efficiently than traditional genetic algorithm and catastrophe adaptive genetic algorithm. By adjusting the time-varying traffic network model parameters in the optimization of its convergence rate, and the ability of disturbance is not large, to show that the algorithm improves the quality of the optimal solution at the same time, the robustness is also improved, showing good solving performance, algorithm design of static and dynamic environment under multimodal transport route optimization problem solving still is efficient and feasible.

5. Conclusions

Based on the characteristics of highway, railway, and waterway, this paper considers the influence of time-varying factors such as traffic jams, the external environment and transport vehicle failure on transport speed. In the time-varying transportation network, the traveling speed of transportation vehicles is affected by the time-varying factors, which leads to the delay of traveling time and the uncertainty of carbon emissions. In the time-varying transportation network, carbon emissions are measured by the actual travel time of the transport vehicle, and the optimization model of the cold chain container multimodal transportation route considering carbon emissions is established under the time-varying network. The improved queen bee genetic algorithm was used to solve the model in two stages, namely, route optimization of cold chain container multimodal transport in a static environment and a real-time optimization of cold chain container multimodal transport in a dynamic environment. The results show that road transport is greatly influenced by the external environment and has adverse effects on the total cost and carbon emissions. Moreover, the multimodal transport process involving road transport is sensitive to the increase in carbon tax value, which can effectively promote the implementation of the policy of "transferring from road to railway mode of transport" on the premise of meeting the time window limit of arriving at the destination of goods. Finally, a numerical example is given to verify that the model can effectively solve the routing optimization problem of cold chain container multimodal transport under the condition of a time-varying traffic network, which provides a theoretical reference for the decision-makers of container transport enterprises. The consideration of the train and water transport route schedules and multi-stage material dynamic distribution optimization will be the direction of further research on cold chain container multimodal transport.

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References

1. Tang, Y.Y.; Dai, W.D.; Chen, S. Container Multimodal Transport Route Planning Considering the Difference in a Multi-node Time Window. *J. Transp. Eng. Inf.* **2020**, *18*, 34–42.
2. Zhang, X.L.; Chen, X.H. Study on Multiple-objective Multimodal Transport route Optimization Under Mixed time Windows. *China Transp. Rev.* **2018**, *40*, 98–105.
3. Li, M. A Study on the Optimization Model of Multimodal Container Transport route in the Yangtze River Economic Belt. *Railw. Transp. Econ.* **2019**, *41*, 94–99.
4. Hu, Y.; Yuhong, S. Multimodal Transportation and Route Optimization of Vehicle Logistics. *J. Transp. Eng. Inf.* **2019**, *17*, 13–19.
5. Liu, Q.; Zhu, X.J.; Zhou, Z.Y.; Yang, Y. Research on Optimization Model of Container Multimodal Transport route of Yangtze River Trunk Line. *J. Wuhan Univ. Technol. Transp. Sci. Eng.* **2019**, *43*, 622–626.
6. Li, M.; Wang, X.; Sun, Q.; Yang, Z. Research on Allocation Strategies of Multimodal Transportation for Emergency Resources Based on Robust Optimization. *J. China Railw. Soc.* **2017**, *39*, 1–9.
7. Li, J.; Yang, B.; Zhu, X.L. Path Optimization of Green Multimodal Transportation under Mixed Uncertainties. *J. Transp. Syst. Eng. Inf. Technol.* **2019**, *19*, 13–21.
8. Liu, S.C.; Zhang, C. Optimization of Urban Cold Chain Transport Routes under Time-Varying Network Conditions. *J. Adv. Transp.* **2021**, *2021*, 1–16. [CrossRef]
9. Liu, D.; Xu, J.C. A Container Transport Planning Approach of China-EU Freight Transport Considering Various Factors. *J. Dalian Jiaotong Univ.* **2000**, *41*, 11–16.
10. Wan, J.; Wei, S. Multi-Objective Multimodal Transportation route Selection Based on Hybrid Algorithm. *J. Tianjin Univ. (Sci. Technol.)* **2019**, *52*, 284–292.
11. Mnif, M.; Sadok, B. Firework algorithm for multi-objective optimization of a multimodal transportation network problem. *Procedia Comput. Sci.* **2017**, *112*, 1670–1682. [CrossRef]
12. Chen, M.F.; Liu, Y.Q.; Song, Y.; Sun, Q.; Cong, C.C. Multimodal Transport Network Optimization Considering Safety Stock under Real-Time Information. *J. Advanced Transp.* **2019**, *2019*, 1–13. [CrossRef]
13. Lozano, A.; Storchi, G. Shortest viable route algorithm in multimodal networks. *Transp. Res. Part A* **2001**, *35*, 225–241.
14. Chen, Y.; Li, Y. Research on Multimodal Transport Optimization Problem Based on Carbon Constraint Policies. In *Proceedings of 4th International Conference on Logistics, Informatics and Service Science*; Springer: Berlin/Heidelberg, Germany, 2014; pp. 339–344.
15. Assadipour, G.; Ke, G.Y.; Verma, M. A toll-based bi-level programming approach to managing hazardous materials shipments over an intermodal transportation network. *Transp. Res. Part D Transp. Environ.* **2016**, *47*, 208–221. [CrossRef]
16. Verga, J.; Silva, R.C.; Yamakami, A. Multimodal transport network problem: Classical and innovative approaches. *Stud. Fuzziness Soft Comput.* **2018**, *2018*, 118–154. [CrossRef]
17. Wang, J.; Wud, W.X. Urban Multimodal transportation system simulation modeling considering carbon emissions. In Proceedings of the 30th Chinese Control and Decision Conference, Shenyang, China, 9–11 June 2018; Northeastern University: Boston, MA, USA, 2018.
18. Nagy, T.; Turanyi, T. Uncertainty of Arrhenius parameters. *Int. J. Chem. Kinet.* **2011**, *43*, 359–378. [CrossRef]
19. Shi, S. *A Study of Vehicle Routing Optimization Problem on Cold Chain Logistics*; Xi'an University of Electronic Science and Technology: Xi'an, China, 2015.
20. Teng, R.E.; Tianyu, L.U.; Zhihua, G.U.; Zhiqing, H.U.; Binbin, J.I.; Lining, X.I. Optimization of urban logistics co-distribution route considering simultaneous pickup and delivery. *Comput. Integr. Manuf. Syst.* **2022**, *28*, 3523.

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